SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA5116 - Cryptography and Network Security

ASSIGNMENT 1

QUESTIONS

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security.

Assignment Title: Evaluate, Design and Defend a Secure Patient-Record Exchange Strategy for a Multi-Hospital Healthcare Network (Total Marks: 50)

Scenario — Confidential Patient Record Exchange

A regional healthcare network links one central hospital with 25 satellite clinics. Clinics send patient records, diagnostic reports, prescriptions and billing data to the central hospital over a public network and receive specialist opinions in return. Personnel hold different clearance levels: Medical Director (Top Secret), Specialist Doctors (Secret), Nurses and Lab Staff (Confidential), Billing and Front-Desk Staff (Restricted), Patients (Public).

The system must:

- Keep diagnoses, especially for sensitive conditions, confidential.
- Detect tampering with prescriptions, dosage and lab values.
- Prevent staff from reading or writing records beyond their clearance.
- Work in rural clinics where staff have limited technical training and hardware is basic.

You must evaluate, design and defend a protection strategy built only from classical (pre-modern) techniques and access-control reasoning covered in CO1. You are expected to judge how strong each technique actually is, not just describe it.

Design Challenge : An insider at a satellite clinic has been copying ciphertexts for months. Hundreds of messages under the same key are now exposed, and many begin with predictable text (e.g., "PATIENT ID", "DIAGNOSIS:", "PRESCRIPTION:"). The network also wants to embed a short report inside medical scan images sent between

hospitals. Re-evaluate your design.

- Encrypt clinical notes (200 to 10,000 characters) using Caesar, monoalphabetic, Vigenère and columnar transposition, and derive each key space. Break at least three schemes by frequency analysis, Kasiski or index of coincidence, and rank the schemes by resistance.
- Extend the alphabet to digits and symbols (mod 36) and encrypt IDs and dosages such as PT-20481, AMOX 500MG using a Hill cipher on 3-character blocks. Prove the key's invertibility, compute one inverse, and recover the key by a known-plaintext attack.

- Apply columnar transposition to a lab table of 20+ rows and test whether a changed cell or a swapped row is detected. Design a classical checksum and show where an attacker defeats it.
- Encrypt a 256×256 or larger medical image with two classical methods (intensity shift mod 256 and pixel permutation) and verify decryption. Compare histograms and entropy, and explain which method leaks the image's structure.
- Hide a short report in an image by LSB embedding and in a text circular by a text method, then extract it. Measure PSNR, capacity and chi-square detectability, and test survival under JPEG compression.
- Decide 12+ access requests using Bell-LaPadula with compartments, including declassification and break-glass cases. Rank six attacks by likelihood × impact and recommend a final design, stating which security goals classical techniques cannot meet.

TASK 1 – Evaluate Threats, Attacks and Risks **4 Marks**

Identify at least six security attacks relevant to the healthcare scenario.

- Classify each as passive or active (eavesdropping, traffic analysis, modification of prescriptions, masquerade as a doctor, replay of old orders, denial of service on the records system).
- Distinguish clearly between threat, vulnerability, attack and risk, with examples from this scenario.
- Rank the attacks by risk using a justified likelihood × impact rating. Consider patient-safety impact, not just privacy.
- Defend your ranking: which attack would you address first, and why?

Evidence Expected: A risk table with reasoned ratings and a justification of the priority order.

TASK 2 – Map and Judge Security Goals **3 Marks**

For each system function (record upload, prescription transmission, specialist consultation), decide which goals matter most: confidentiality integrity availability authentication non-repudiation

- Judge which goals classical ciphers can and cannot deliver.
- Justify why a correctly decrypted message does not prove it came from the right doctor, or that its dosage was not altered.

Evidence Expected: A goal-to-function analysis with an evaluation of the gaps classical techniques leave.

TASK 3 – Apply and Critique the Bell-LaPadula Model **5 Marks**

- Define the security levels and subject/object assignments for the network.
- Apply the simple security property (no read up) and the *-property (no write down) to at least five access requests, including two that must be denied. Show the reasoning for each.

- Evaluate the model critically in a medical setting: what happens in an emergency when a nurse needs a Secret-level allergy record? How does the model handle a doctor who must write a note to a lower-level billing file? Discuss tranquility and trusted subjects.
- Judge the model's limitations (no integrity guarantees, covert channels, rigidity in emergencies) and argue whether Bell-LaPadula alone suits healthcare.

Evidence Expected: Worked access decisions, a justified critique, and a clear verdict on the model's suitability.

TASK 4 – Design the Cryptographic Strategies **4 Marks**

Select at least three meaningfully different classical schemes from: Caesar, monoalphabetic substitution, Vigenère (polyalphabetic), Hill cipher, a transposition cipher. For each, provide:

- The encryption and decryption algorithm (pseudocode or formula) and the key definition.
- A worked example on a 20+ character message from the scenario.
- A justification for including it in your comparison.

Evidence Expected: Three distinct schemes with algorithms, worked examples and justified selection.

TASK 5 – Mathematical Analysis of the Schemes **5 Marks**

- Derive the key-space size of each scheme (e.g., 25, 26!, 26^m, number of invertible n×n matrices mod 26, n! permutations).
- For Hill: show the conditions under which K^{-} mod 26 exists (gcd(det K, 26) = 1), and compute one inverse by hand.
- Express each encryption as a modular function (e.g., C = (aP + b) mod 26 for an affine variant, C = KP mod 26 for Hill).
- Judge whether a large key space guarantees strength. Support your answer with the monoalphabetic example.

Evidence Expected: Correct derivations with a reasoned judgment on key space versus real security.

TASK 6 – Evaluate Cryptanalytic Resistance **4 Marks**

Attack each chosen scheme using a technique suited to it:

- Brute force, frequency analysis, Kasiski examination / index of coincidence, known-plaintext attack on Hill, digram analysis.
- Explain why each attack succeeds or fails, in terms of letter-frequency preservation, key repetition and linearity.
- Rank the schemes by resistance and justify the ranking with evidence.

Evidence Expected: Worked attacks with reasoning, and a ranking of schemes supported by the results.

TASK 7 – Assess Transposition and Product Ciphers **3 Marks**

- Implement a columnar (or rail-fence) transposition and explain how it diffuses plaintext structure.
- Evaluate whether *substitution followed by transposition* is stronger than either alone. Justify with letter-frequency evidence and an attack attempt.
- Judge whether the added complexity is practical for rural clinic staff doing manual or low-resource encryption.

Evidence Expected: A reasoned evaluation of the combined scheme against its components.

TASK 8 – Evaluate Steganography **3 Marks**

Design a steganographic method (e.g., LSB embedding in a scan image, or text-based hiding) for carrying a short report inside a medical image.

- Compare it with encryption on capacity, imperceptibility, robustness and detectability.
- Evaluate the risk specific to medical imaging: could embedding alter pixel data enough to affect a diagnosis? Judge the trade-off.
- Judge when steganography is appropriate, and argue why hiding is not a substitute for encrypting. Evaluate an encrypt-then-embed approach.

Evidence Expected: A criteria-based comparison with a justified conclusion.

TASK 9 – Implement and Test **4 Marks**

Implement all chosen schemes, the cryptanalysis tools and (where feasible) the steganography method, in a language of your choice.

- Provide correctness tests: decrypt(encrypt(m)) = m, including edge cases (digits in patient IDs, punctuation, non-multiple block length for Hill).
- Use identical plaintexts of 100, 1,000, 10,000 and 100,000 characters across schemes.
- State the testing procedure clearly so it can be reproduced.

Evidence Expected: Working code with a documented, reproducible test procedure

TASK 10 – Record and Interpret Experimental Evidence **3 Marks**

Present data in tables and graphs for:

- Letter-frequency histograms of plaintext versus each ciphertext.
- Index of coincidence per scheme.
- Encryption/decryption time against message size.
- Attack success rate against ciphertext length.

Evaluate what the data shows. Do not just present it.

Evidence Expected: Clear, comparable results with interpretation tied to Tasks 5 and 6.

TASK 11 – Recommend and Defend the Final Design **4 Marks**

Recommend one scheme, or a layered combination, together with the Bell-LaPadula policy.

- Justify against the alternatives using your mathematical and experimental evidence.
- State the residual risks and limitations honestly. Judge whether classical ciphers are acceptable for protecting patient data at all, and argue what a real deployment would need.
- Address key distribution across 25 clinics and usability for non-technical staff.

Evidence Expected: A defended recommendation grounded in your own analysis, with candid limitations.

DAY- 4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design **8 Marks**

Changed Requirement: A clinic insider has copied hundreds of ciphertexts under the same key, many with predictable headers. The network also wants reports hidden inside medical scan images.

You may RETAIN → MODIFY → or REDESIGN your original solution.

A. Impact Analysis (2 Marks)

- What has changed in the threat model (insider versus external attacker)?
- Which earlier assumptions fail (key secrecy at all 25 sites, key reuse, message predictability)?
- Why does a high volume of known-format messages change the risk ranking?

B. Evaluation of the Original Design (2 Marks)

- Test your recommended scheme against the new attacker capabilities (known-plaintext, repeated-key attacks).
- Support the evaluation with a demonstration or calculation.
- Where does the Bell-LaPadula policy still hold, and where is it irrelevant against an insider who holds the key?

C. Adaptation or Redesign (2 Marks)

- Decide to retain, modify or redesign. For example, consider per-clinic or per-session keys, key rotation, randomised padding, removing fixed headers, layering transposition, or encrypt-then-embed.
- Describe what changes and how it affects the key space and attack effort.

D. Evidence-Based Defence (2 Marks)

Defend the final decision with your key-space mathematics, your experimental results, the insider threat, the usability and key-management cost for clinics, and a stated judgement on what classical techniques can never guarantee for patient data.

MARK DISTRIBUTION

Task	Marks
Task 1 – Threats, Attacks and Risk Evaluation	4
Task 2 – Security Goals Mapping	3
Task 3 – Bell-LaPadula Application and Critique	5
Task 4 – Cipher Design	4
Task 5 – Mathematical Analysis	5
Task 6 – Cryptanalytic Resistance	4
Task 7 – Transposition / Product Ciphers	3
Task 8 – Steganography Evaluation	3
Task 9 – Implementation & Testing	4
Task 10 – Experimental Evidence	3
Task 11 – Final Design Recommendation	4
Subtotal	**42**
Task 12 – Day-4 Design Challenge	**8**
TOTAL	**50**

CSA5116 – Cryptography and Network Security
ASSIGNMENT 1 (CO1)

Evaluate, Design and Defend a Secure Patient-Record Exchange Strategy for a Multi-Hospital Healthcare Network

Conventions and reproducibility. Classical ciphers work on the letters A–Z after upper-casing and deleting spaces, digits and punctuation; the Hill cipher works on A–Z and 0–9 (mod 36); the padding letter is X. Every number in this report was produced by the accompanying Python programs (crypto_lab.py, run_experiments.py, make_figs.py). Clinical text is generated from 20 realistic note sentences with a fixed seed; the attacker's reference statistics come from independent English text (open-source licence texts), never from the clinic plaintexts. The scan is a synthetic 256×256 head phantom, so no real patient data was used. The answers also cover the mark-sheet items: cipher design and pseudocode (Task 4), key analysis (5), cryptanalysis and comparison (6, 11), implementation and results (9, 10) and re-evaluation 12A–12D (Task 12).

TASK 1 – Evaluate Threats, Attacks and Risks

Term	Meaning	Example in this network
Threat	A potential cause of harm (an actor or event).	A dissatisfied clinic employee who can reach the public network.
Vulnerability	A weakness that a threat can use.	One key shared by all 25 clinics; fixed headers such as "PATIENT ID"; no check that an order really came from the doctor.
Attack	An action that exploits a vulnerability.	Changing "AMOX 500MG" to "AMOX 900MG" in transit, or collecting ciphertexts and running frequency analysis.
Risk	Likelihood × impact of an attack succeeding.	Dosage modification: likelihood 4 × impact 5 (possible death) = 20.

Risk table. Likelihood (L) and impact (I) are rated 1–5; impact is weighted towards patient safety, not only privacy. The last numeric column re-rates each attack after the Day-4 insider change (Task 12A).

#	Attack	Type	L	I	Risk	Day-4	Justification
1	Modification of prescription, dosage or lab values	Active	4	5	20	25	Silent; one changed character changes the dose; classical ciphers add no integrity check; direct harm to the patient.
2	Masquerade as a doctor (forged order)	Active	3	5	15	20	A shared key lets any key holder produce a valid-looking order; wrong therapy possible.
3	Eavesdropping / ciphertext collection	Passive	5	3	15	20	Trivial on a public network; harm is loss of privacy of diagnoses; grows with the stored ciphertext volume.
4	Replay of old orders	Active	3	4	12	12	A recorded valid order re-sent causes duplicate dosing; no counter or time stamp exists.
5	Denial of service on the records system	Active	3	4	12	12	Emergency care without allergy history; rural hardware has no redundancy.
6	Traffic analysis (who, when, how much)	Passive	4	2	8	8	Reveals referral patterns (for example oncology traffic) but no content.
7	Insider key theft + known-plaintext cryptanalysis (added after Day 4)	Passive, then active	2	5	10	25	The first model assumed outsiders; with cribs and months of ciphertext it becomes the root cause of rows 1–3.

Defence of the ranking and first priority. I would address **row 1 (modification of prescriptions and lab values) first**. It has the highest score once patient safety is weighted, it is silent (nobody notices until harm occurs), and classical ciphers cannot stop it, so the countermeasure has to be added deliberately: redundancy checks, a counter, and out-of-band confirmation of high-risk orders, all cheap enough for rural clinics. Eavesdropping has the highest likelihood (5) but a lower impact (3), so it ranks third. Masquerade is second because it exploits the same missing authentication. Where scores tie, the attack that can physically harm a patient ranks higher.

TASK 2 – Map and Judge Security Goals

Function	Confid.	Integrity	Avail.	Authent.	Non-rep.	What classical techniques deliver
Record upload	High (diagnoses)	High	Medium	High	Medium	Weak confidentiality only; no proof of origin; stored ciphertext can be re-sent.
Prescription transmission	Medium	Critical	High	High	High	Nothing for integrity, authentication or non-repudiation; a changed dose still decrypts cleanly (Task 7 test).
Specialist consultation	High	High	High (emergency)	High	Medium	Confidentiality only; an opinion can be forged by anybody who holds the shared key.

Judgement. Classical ciphers can contribute only to confidentiality, and only weakly (Tasks 5–6). They cannot deliver integrity, authentication, non-repudiation or availability. For prescription transmission, where the most important goal is integrity, they give almost nothing.

Why a correctly decrypted message proves nothing. (1) *Origin:* with a symmetric key shared by 25 clinics, successful decryption only shows that the sender knew the key, so any key holder can write "from Dr. Rao" and the sender can later deny it. (2) *Dosage:* Caesar, Vigenère and Hill are linear, so adding δ to a ciphertext letter adds δ to the plaintext letter; a transposition moves letters without changing them. In my lab-table test (Task 7) altering one ciphertext character changed exactly one plaintext character and decryption reported no error, because plain text has no built-in redundancy that would make a damaged message look invalid. (3) *Freshness:* an old valid ciphertext decrypts correctly forever (replay).

TASK 3 – Apply and Critique the Bell-LaPadula Model

Levels and labels. Levels: Public (0) = patients; Restricted (1) = billing and front desk; Confidential (2) = nurses and lab staff; Secret (3) = specialist doctors; Top Secret (4) = Medical Director. Compartments (need-to-know categories): GEN, ONC, PSY, HIV, BILL. A label (L, C) dominates (L′, C′) when L ≥ L′ and C ⊇ C′. The **simple security property** allows a read only if the subject's label dominates the object's (no read up). The **star property** allows a write only if the object's label dominates the subject's (no write down).

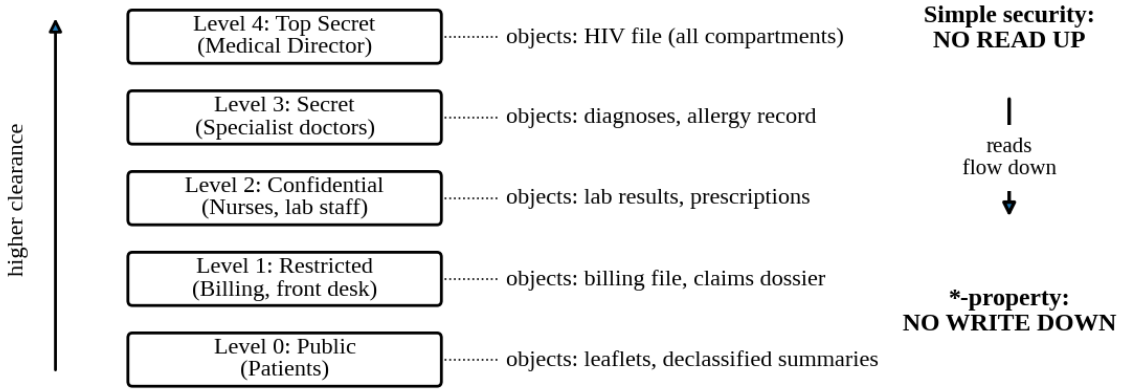


Figure 1. Bell-LaPadula levels, subjects and objects: reads go down, writes go up.

Subjects: Medical Director (TopSecret, all compartments); Dr. Rao (Secret, {GEN, ONC}); Dr. Mehta (Secret, {GEN, PSY}); Nurse Anita and Lab technician Ravi (Confidential, {GEN}); Billing clerk Sita (Restricted, {BILL}); Patient (Public, {}). **Objects:** lab result (Confidential, {GEN}); allergy record (Secret, {GEN}); oncology diagnosis (Secret, {GEN, ONC}); HIV status (Secret, {HIV}); psychiatric note (Secret, {GEN, PSY}); billing file (Restricted, {BILL}); claims dossier (Confidential, {BILL}); public leaflet (Public, {}); discharge summary (Confidential, {GEN}, later declassified to Public). The 16 decisions below were computed by the program (blp() in crypto_lab.py) and checked by hand.

#	Subject [level/compartments]	Op	Object [label]	Decision	Reasoning
1	Nurse Anita [Confidential/{GEN}]	read	Lab result [Confidential/{GEN}]	ALLOW	Confidential ≥ Confidential and {GEN} ⊇ {GEN}: simple security satisfied.
2	Nurse Anita [Confidential/{GEN}]	read	Allergy record [Secret/{GEN}]	DENY	Confidential < Secret: read up is forbidden. Emergency path: see request 16.
3	Dr Rao (oncology) [Secret/{GEN,ONC}]	read	Oncology diagnosis [Secret/{GEN,ONC}]	ALLOW	Secret ≥ Secret and {GEN, ONC} ⊇ {GEN, ONC}: allowed.
4	Dr Rao (oncology) [Secret/{GEN,ONC}]	read	HIV status [Secret/{HIV}]	DENY	Level is equal but {GEN, ONC} does not contain {HIV}: no need-to-know, so denied despite the same level.
5	Dr Rao (oncology) [Secret/{GEN,ONC}]	write	Oncology diagnosis [Secret/{GEN,ONC}]	ALLOW	Object label (Secret, {GEN, ONC}) dominates the subject: star property satisfied (same-level write).
6	Dr Rao (oncology) [Secret/{GEN,ONC}]	write	Billing file [Restricted/{BILL}]	DENY	Object is lower (Restricted): write down would let diagnosis details leak into a low file, so denied.
7	Billing clerk Sita [Restricted/{BILL}]	read	Lab result [Confidential/{GEN}]	DENY	Restricted < Confidential: read up denied.
8	Billing clerk Sita [Restricted/{BILL}]	write	Claims dossier [Confidential/{BILL}]	ALLOW	Object (Confidential, {BILL}) dominates the subject, so the write is legal; but this blind write-up is how BLP lets a low user corrupt higher data (integrity gap).
9	Medical Director [TopSecret/{BILL,GEN,HIV,ONC,PSY}]	read	HIV status [Secret/{HIV}]	ALLOW	TopSecret ≥ Secret and all compartments held: allowed.
10	Medical Director [TopSecret/{BILL,GEN,HIV,ONC,PSY}]	write	Public leaflet [Public/{}]	DENY	Write down from TopSecret to Public: forbidden, even for the Director (prevents a Trojan horse leaking data).
11	Patient [Public/{}]	read	Discharge summary (Conf.) [Confidential/{GEN}]	DENY	Public < Confidential: patient cannot read the summary while it is Confidential.
12	Patient [Public/{}]	read	Discharge summary (declassified) [Public/{}]	ALLOW	After a trusted officer declassifies the summary to Public, Public ≥ Public: allowed (a label change, see tranquility below).
13	Dr Mehta (psychiatry) [Secret/{GEN,PSY}]	read	Psychiatric note [Secret/{GEN,PSY}]	ALLOW	Secret ≥ Secret and {GEN, PSY} ⊇ {GEN, PSY}: allowed.
14	Dr Mehta (psychiatry) [Secret/{GEN,PSY}]	read	Oncology diagnosis [Secret/{GEN,ONC}]	DENY	{GEN, PSY} does not contain {ONC}: compartment check fails, denied.
15	Lab tech Ravi [Confidential/{GEN}]	write	Lab result [Confidential/{GEN}]	ALLOW	Same label on subject and object: write allowed.

#	Subject [level/compartments]	Op	Object [label]	Decision	Reasoning
16	Nurse Anita [Confidential/{GEN}]	read	Allergy record [Secret/{GEN}] – BREAK-GLASS	ALLOW*	Outside plain BLP: emergency flag, reason typed, second person confirms; read-only grant for 30 minutes; every access logged and reviewed afterwards.

Critical evaluation in a medical setting. *Emergency:* a nurse who needs a Secret allergy record in an anaphylaxis emergency is refused by strict BLP (request 2), and delay can kill; the policy therefore needs an audited break-glass override (request 16), which BLP itself does not contain. *Doctor writing to a billing file:* the star property forbids it (request 6) because the doctor's label is higher than the file's. Solutions are a **trusted subject** (a guard process exempt from the star property that extracts only billing codes and diagnoses-free text), or a sanitised summary that is declassified. *Tranquility:* strong tranquility (labels never change) would make it impossible to release a discharge summary to the patient (requests 11–12); weak tranquility lets labels change only under rules and only through trusted officers. Trusted subjects and declassifiers lie outside the model's guarantees and enlarge the trusted computing base, so they must be few, small and audited.

Limitations and verdict. (1) BLP gives no integrity: write-up is allowed, so a clerk can overwrite a higher record (request 8), and nothing protects dosage values. (2) Covert channels remain, for example the error message "file exists but access denied" or timing differences reveal a Secret file. (3) It is rigid in emergencies and label management with compartments becomes heavy. (4) It controls access inside the reference monitor only; it cannot constrain a person who can decrypt the data outside the monitor (Task 12B). **Verdict:** BLP alone is not suitable for healthcare. It is a good confidentiality core, but must be combined with an integrity model (Biba or Clark-Wilson), role-based rules, break-glass and auditing.

TASK 4 – Design the Cryptographic Strategies

Five distinct schemes are used. Letter values: A = 0 … Z = 25; for Hill, 0–9 = 26–35.

Scheme	Key (used here)	Encryption / decryption	Why included
Caesar	shift k = 7 (1–25)	$C = (P + k) \bmod 26$; $P = (C - k) \bmod 26$	Baseline; shows total collapse under brute force.
Monoalphabetic	permutation π from keyword HOSPITAL: HOSPITALBCDEFGJKMNQRUVWXYZ	$C = \pi(P)$; $P = \pi^{-1}(C)$	Huge key space (26!) yet weak; tests "big key = strong".
Vigenère	keyword HEALTH (m = 6)	$C_i = (P_i + k_{i \bmod m}) \bmod 26$; $P_i = (C_i - k_{i \bmod m}) \bmod 26$	Polyalphabetic; flattens frequencies; key-length attacks.
Columnar transposition	keyword HOSPITAL (m = 8); column order by alphabetical rank	Write plaintext row-wise under the key (pad X); read columns in key order. Decrypt: refill columns in key order, read rows.	Permutation; diffusion of structure; letter frequencies unchanged.
Hill, n = 3, mod 36	K = [[1,2,8],[5,7,9],[2,5,6]], det = 61	$C = K \cdot P \bmod 36$; $P = K^{-1} \cdot C \bmod 36$	Linear algebra; works on digits (IDs, dosages); linear, so weak to known plaintext.

Pseudocode (with input validation).

Procedure	Steps
VIGENÈRE-ENC(text, key)	1 text ← upper-case(text) with all non-letters removed; if key is empty or has a non-letter → error 2 for i = 0 … n−1: C[i] ← (P[i] + K[i mod m]) mod 26 3 return C (decryption: step 2 with "−" instead of "+")
COLUMNAR-ENC(text, key)	1 clean text; m ← length(key); while n mod m ≠ 0: append X 2 order ← column indices sorted by (key letter, index) 3 for col in order: append P[col], P[col+m], P[col+2m], … to C; return C 4 decryption: rows ← n/m; cut C into m blocks of length rows; block t is column order[t]; read row by row
HILL-ENC(text, K)	1 keep only A–Z and 0–9; pad X until length is a multiple of 3 2 if gcd(det K mod 36, 36) ≠ 1 → error "key not invertible" 3 for each block P of 3 characters (column vector): C ← K·P mod 36 4 decryption uses $K^{-1} = (\det K)^{-1} \cdot \mathrm{adj}(K) \bmod 36$

Worked example. Message: "Prescribe amoxicillin twice daily" → PRESCRIBEAMOXICILLINTWICEDAILY (30 letters). Vigenère check for the first six letters: P+H = 15+7 = 22 → W; R+E = 17+4 = 21 → V; E+A = 4 → E; S+L = 18+11 = 29 → 3 → D; C+T = 2+19 = 21 → V; R+H = 17+7 = 24 → Y, giving WVEDVY.

Scheme / key	Ciphertext produced	Decryption result
Caesar, k = 7	WYLZJYPILHTVEPJPSSPUADPJLKHPSF	original recovered ✓
Monoalphabetic, HOSPITAL	KNIQSNBOIHFJXBSBEEBGRWBSIPHBEY	original recovered ✓
Vigenère, HEALTH	WVEDVYPFELFVEMCTESPRTHBJLHATEF	original recovered ✓
Columnar, HOSPITAL (32 letters: 2 pad X)	ICIXPELECXTLBICXRALDSONIEMIARIWY	original + XX (pad removed) ✓
Hill: PT-20481 → PT20481 + XX	Z2F88KF9T	PT20481XX ✓
Hill: AMOX 500MG → AMOX500MG	24AF070QE	AMOX500MG ✓

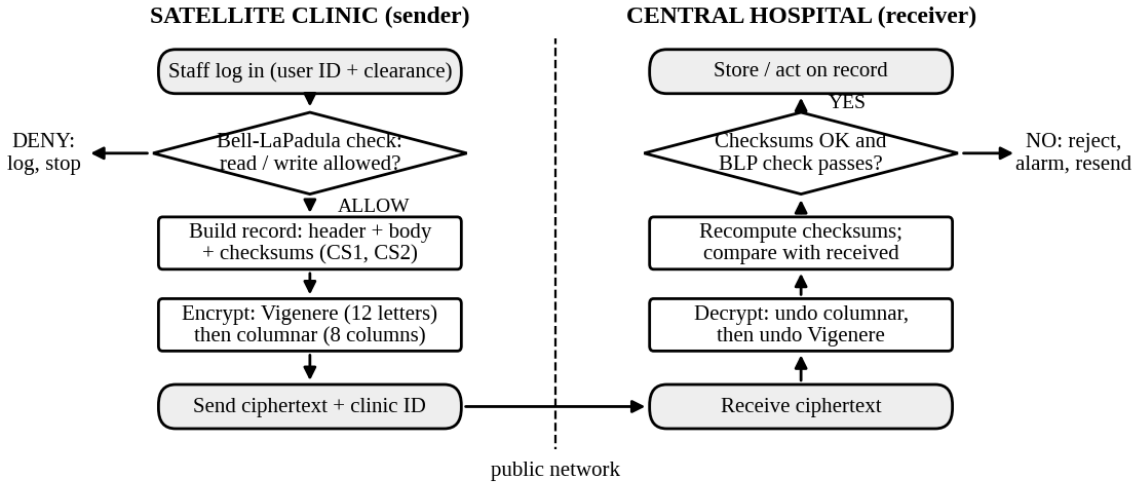


Figure 2. Original protocol: BLP check, record build with checksums, product encryption, public network, verification at the hospital.

TASK 5 – Mathematical Analysis of the Schemes

Scheme	Key-space derivation	Size	Bits
Caesar	26 shifts − identity	25	4.6
Affine (variant)	$\varphi(26) \cdot 26 = 12 \times 26$	312	8.3
Monoalphabetic	26! permutations	4.03×10^{26}	88.4
Vigenère, m = 6 / m = 12	26^m	3.09×10^{8} / 9.54×10^{16}	28.2 / 56.4
Columnar, m = 8 / m = 12	m! orderings	40,320 / 479,001,600	15.3 / 28.8
Hill 2×2 mod 26	$\lvert GL_2(Z_2)\rvert \cdot \lvert GL_2(Z_{13})\rvert = 6 \times 26{,}208$	157,248	17.3
Hill 3×3 mod 26	$168 \times (13^3-1)(13^3-13)(13^3-13^2)$	1.63×10^{12}	40.6
Hill 3×3 mod 36 (used)	$\lvert GL_3(Z_4)\rvert \cdot \lvert GL_3(Z_9)\rvert = (2^9 \cdot 168)(3^9 \cdot 11{,}232)$	1.90×10^{13}	44.1
Product Vigenère-12 + columnar-8	$26^{12} \cdot 8!$	3.85×10^{21}	71.7
Product mono + columnar-8	$26! \cdot 8!$	1.63×10^{31}	103.7

Modular functions. Caesar: $C \equiv P + k \pmod{26}$. Affine: $C \equiv aP + b \pmod{26}$, gcd(a, 26) = 1. Monoalphabetic: $C = \pi(P)$. Vigenère: $C_i \equiv P_i + k_{i \bmod m} \pmod{26}$. Hill: $\mathbf{C \equiv K \cdot P \pmod{36}}$. Columnar: $C = \Pi \cdot P$, where Π is an n×n permutation matrix (one 1 in each row and column).

Hill invertibility. K^{-1} exists modulo m exactly when gcd(det K, m) = 1, because only then det K has an inverse modulo m. For m = 36 = 4·9 the determinant must be odd and not a multiple of 3; for m = 26 it must be odd and not a multiple of 13. For K = [[1,2,8],[5,7,9],[2,5,6]]: det = 1(7·6−9·5) − 2(5·6−9·2) + 8(5·5−7·2) = −3 − 24 + 88 = **61 ≡ 25 (mod 36)**, gcd(25, 36) = 1, so K is invertible, and $25^{-1} \equiv$ **13** because 25·13 = 325 = 9·36 + 1. The adjugate (transposed cofactor matrix) is [[−3, 28, −38], [−12, −10, 31], [11, −1, −3]], so $K^{-1} = 13 \cdot \text{adj}(K) \bmod 36$ = **[[33, 4, 10], [24, 14, 7], [35, 23, 33]]**; $K \cdot K^{-1} = I \pmod{36}$ was verified. A non-example, [[2,4,6],[1,3,5],[7,8,9]], has det = 0, gcd = 36, and the program rejects it.

Does a large key space guarantee strength? No. The monoalphabetic cipher has 26! = 4.03×10^{26} keys (88.4 bits), far beyond exhaustive search, yet it preserves every letter frequency (its index of coincidence stays 0.0659, identical to the plaintext) and every digram pattern. A hill-climbing attack that needs only about 10^4 trial key swaps ($\approx 2^{14}$) recovered at least 90% of the letters in 95% of trials with 500 letters of ciphertext (Fig. 6). Key space is necessary but not sufficient: strength depends on how much statistical structure of the plaintext survives encryption, on key reuse, and on linearity.

TASK 6 – Evaluate Cryptanalytic Resistance

Scheme	Attack used	Why it succeeds or fails	Measured result (40 trials per length)
Caesar	Brute force over 25 keys, chi-square against English frequencies	A shift preserves the shape of the frequency profile; only 25 keys.	100% at every length from 30 letters.
Monoalphabetic	Frequency analysis, then hill-climbing on digram statistics	One-to-one letter mapping preserves frequencies and digrams. Fails on short text because the statistics are noisy.	≥90% of letters correct: 0% at 30–60, 2.5% at 120, 30% at 250, 95% at 500, 97.5% at 1,000.
Vigenère (m = 6)	Kasiski spacing + index of coincidence for key length, then per-column chi-square	Repeated key makes repeated plaintext fragments recur at multiples of m; every column is a Caesar cipher. Fails when text is too short.	Key recovered: 0%, 10%, 40%, 90%, 100%, 92.5% for 30…1,000 letters.
Columnar (m = 8)	Digram scoring over all 8! = 40,320 column orders	Letter frequencies unchanged (IC stays 0.0659) but digrams are broken; the small key space allows exhaustive search.	Exact plaintext: 2.5%, 42.5%, 80%, 97.5%, 100%, 100%.
Hill, 3×3 mod 36	Known-plaintext: $K = C \cdot P^{-1}$	Encryption is linear, so three independent known blocks give the key. Ciphertext-only is much harder (IC 0.0286 ≈ 1/36).	Key recovered exactly from PAT, IEN, GNO → X4A, MFG, A5R; success independent of length.

Evidence for the Vigenère attack. Average IC of the columns for key length m: 0.0651 at m = 6 and 0.0650 at m = 12, against about 0.045 for every other m (the guess 6 is therefore correct). Kasiski spacing votes concentrate on the divisors 2 (655), 3 (634), 6 (618) and 12 (354), pointing to period 6. The few failures at 1,000 letters come from my threshold rule choosing a factor of the true length, not from a limit of the method. For Hill, a triple of known blocks must give an invertible matrix mod 36: only 31 of the 120 triples I formed from ten header blocks did (PAT, IEN, TID did not), but an insider with hundreds of messages always finds one.

Ranking by resistance (strongest first), justified by the data: (1) **Monoalphabetic**: needs about 500 letters, and 88.4 key bits. (2) **Vigenère**: about 250 letters for m = 6, longer keys need more text, but one 12-letter crib reveals the whole key (Task 12). (3) **Columnar**: about 250 letters, only 15.3 key bits. (4) **Hill**: resists frequency analysis best, but nine known characters break it, which predictable headers supply. (5) **Caesar**: broken with a handful of letters. The product of two of these (Task 7) outranks all single schemes against automated attacks.

TASK 7 – Assess Transposition and Product Ciphers

Diffusion by columnar transposition. With m = 8 columns, plaintext letters that were neighbours end up about n/8 positions apart, so digram structure is broken while single-letter statistics are untouched: the share of the ten most common English digrams (TH, HE, IN …) drops from 16.4% to 7.8%, but the IC stays 0.0668. A transposition is therefore recognisable as one, and it is attackable (Task 6).

Measure (1,600 letters)	Mono only	Columnar only	Mono + columnar
Index of coincidence	0.0668	0.0668	0.0668
Letters recovered by direct digram hill-climb	99.3%	not applicable	5.6%
Column-order search by digram-IC (invariant under substitution), then hill-climb	not applicable	not applicable	99.3% in 0.8 s
Key bits	88.4	15.3	103.7

Is substitution followed by transposition stronger? Yes against the standard attack: the direct attack that recovers 99.3% from the substitution alone recovers only 5.6% (roughly chance) from the product, because transposition removed the digram statistics that the attack scores. But it is not much stronger in effort: the attacker simply splits the problem. Digram frequency distributions are unchanged by a letter substitution, so scoring all 8! column orders by the sharpness of the digram distribution found the right order, and then the substitution fell as before (99.3% of letters in 0.8 s). The product adds 15.3 key bits but, with divide-and-conquer, only about 2^{15} extra work.

Practicality for rural staff (my estimate, not measured). Substituting 200 letters from a printed key card and writing a transposition grid takes roughly 15–25 minutes, and one slip in the grid garbles a whole column. Computers do it in under 0.2 ms (Task 10), so the product is practical only with a small low-cost program; manual use invites errors and key-card mistakes.

Lab-table integrity test. A table of 24 rows × 8 characters (test code, 4-digit value, status) was encrypted with columnar transposition (key FGHABCDE). Checksum CS1 = sum of character values mod 36; CS2 = Σ (position × value) mod 97; clean table: CS1 = 12, CS2 = 54.

Attack on the ciphertext	Effect after decryption	CS1	CS2	Detected?
One ciphertext character changed (no key)	exactly 1 plaintext character changed (row 5), no decryption error	11	15	Yes, by both checksums; transposition alone: no
Rows 6 and 18 swapped by exchanging positions 6 and 18 in each of the 8 column blocks (no key needed)	valid table with two rows exchanged	12	62	CS1 misses it (sum unchanged); CS2 detects it
Compensated edit: HGB 0138 → 0148 (+1 at position 6) and BIL 0007 → 0006 (−1 at position 103)	valid table with two altered results	12	54	Neither checksum detects it

Where the attacker defeats the checksum. A checksum of this kind is linear and unkeyed: an insider who knows the formula chooses changes d and e at positions a and b with $(a+1)d + (b+1)e \equiv 0 \pmod{97}$ and $d + e \equiv 0 \pmod{36}$, and both values stay unchanged (found by search above). A keyed message authentication code or signature is the real fix, and it is not a classical technique.

TASK 8 – Evaluate Steganography

Image encryption (256×256, 8-bit). Method 1: intensity shift C = (P + 97) mod 256. Method 2: keyed pixel permutation of all 65,536 positions. Both decrypt exactly (verified bit by bit).

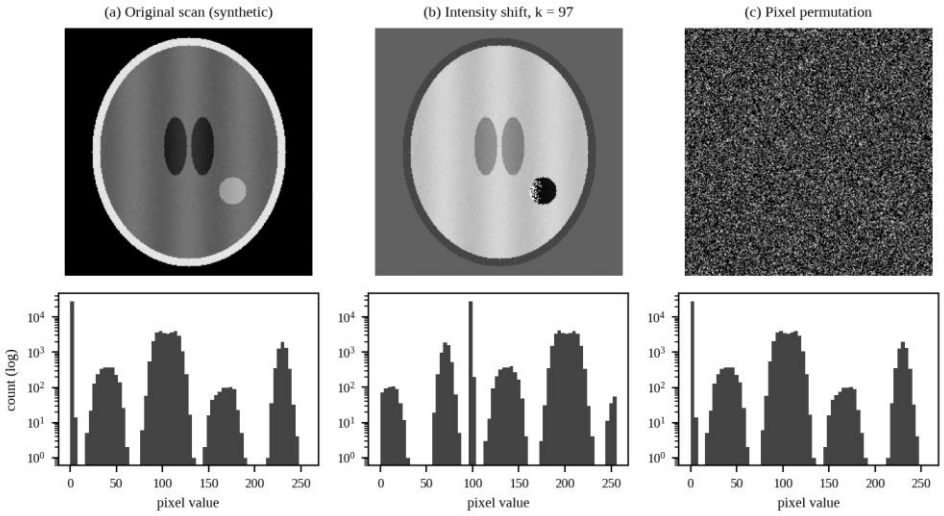


Figure 3. Scan, intensity-shifted image and permuted image with their histograms (log scale).

Measure	Original	Intensity shift	Pixel permutation
Entropy (bits/pixel)	4.817	4.817	4.817
Adjacent-pixel correlation	0.945	0.957	0.003
Histogram distance from original (L1)	–	123174	0
Edge-map correlation with original	1.000	0.477	-0.004
Key space	–	256	$65{,}536! \approx 10^{287{,}194}$

Which leaks structure? The **intensity shift leaks the image's structure**: skull, ventricles and lesion remain visible, neighbouring pixels stay correlated (0.957), and the key is recovered from one histogram peak (the black background peak moved by 97, the true key). The **permutation destroys spatial structure** (correlation 0.003) but its histogram is identical to the original, so it leaks the intensity distribution (for example how much bone and brain are present). The entropy of both is unchanged at 4.817 bits, far from the 8 bits of a good cipher, so neither is secure; a single known image also reveals the permutation.

Steganographic method. The 81-character report "PATIENT PT20481 DIAGNOSIS … REVIEW 1 WEEK" (648 bits) is hidden by LSB replacement at keyed pseudo-random positions, preferably in background pixels outside the head, and extracted with the same key. Capacity is 1 bit per pixel: 65,536 bits (8,192 bytes); background pixels alone give 28,483 bits. Text method: a hospital circular of 123 words carries one bit per gap between words (single space = 0, double space = 1), 5 bits per letter.

Embedded payload	Bits	PSNR (dB)	MSE inside head	χ^2 stat.	p (embedded)	Report extracted
Report, keyed random pixels (whole image)	648	71.73	0.0045	38.8	0.015	yes, exactly
Report, keyed random pixels (background only)	648	71.92	0	27.4	0	yes, exactly
Random (encrypted-like) payload, 10% of pixels	6553	61.12	0.0138	12.1	0.998	n/a (random bits)
Random (encrypted-like) payload, 50% of pixels	32768	54.14	0.2514	32.8	0.998	n/a (random bits)
Random (encrypted-like) payload, 100% of pixels	65536	51.13	0.5026	23.4	1	n/a (random bits)

Detectability and survival. The pairs-of-values chi-square test marks a region as embedded when p is close to 1. Payloads of 10%, 50% and 100% of the pixels give p = 0.998–1.000, whereas the untouched cover gives p = 0.000 at the same positions, so large payloads are detected. The 81-character report (about 1% of capacity) stays below detection ($p \leq 0.015$). *Survival:* PNG keeps the report (bit error rate 0%); JPEG at quality 95, 75 and 50 destroys it (bit error rates 34.1%, 35.5%, 35.5%). The text method extracted "PTTWENTYAMOXFIVEHUNDRED" correctly, but after the whitespace was collapsed (as e-mail, web pages or word processors do) the output became "AAAAAAAA…": all information lost. Also 43.4% of the gaps are double spaces, which is visible in a plain-text view.

Criterion	LSB steganography	Encryption (classical)
Capacity	1 bit/pixel: 8,192 bytes in 256×256; 3,560 bytes background-only	Unlimited; ciphertext as long as the message
Imperceptibility	PSNR 51–72 dB; invisible; hides that a message exists	Obvious cipher text; attracts attention
Robustness	Lost under JPEG (34–36% bit errors), resizing, re-saving	Survives any lossless copy; not format-dependent
Detectability	Chi-square flags payloads ≥10%; 1% payload not flagged	Always recognisable as encrypted but unreadable without key
Secrecy rests on	Secrecy of the method and position key	Secrecy of the key only (Kerckhoffs)

Medical-imaging risk and trade-off. Each LSB change alters a pixel by at most one grey level (0.4% of an 8-bit range): the whole-image keyed embedding gave PSNR 71.7 dB and mean squared error 0.0045 inside the head, while embedding in background pixels only gave

an exact 0.0000 inside the head. A change of one grey level is far below diagnostic thresholds, but the image's hash and DICOM integrity change, and a very subtle lesion near the display threshold could in principle be affected, so my rule is: embed only outside the region of interest, use lossless PNG, and never embed in a diagnostic study without agreed policy. **When is steganography appropriate?** When the existence of a communication must be hidden or there is no metadata channel. **Hiding is not encrypting:** once the method is known the report can be read by anybody (the extraction needs no secret if the positions are fixed), so the safe approach is **encrypt-then-embed** (Fig. 7): the extracted bits are only ciphertext, so detection does not reveal the report. The cost is a higher detectability (random bits equalise histogram pairs) and a 5-bit-per-letter expansion.

TASK 9 – Implement and Test

Language and files. Python 3 with numpy, scipy and Pillow. crypto_lab.py contains the Caesar, monoalphabetic, Vigenère, columnar and Hill ciphers (3×3, mod 36, with adjugate-based inverse), the attacks (chi-square Caesar, bigram hill-climbing, Kasiski and IC, column-order search, Hill known-plaintext) and the Bell-LaPadula engine; run_experiments.py runs every experiment and writes results.json; make_figs.py draws every figure. Steganography and image-encryption code is inside run_experiments.py.

Reproducible procedure. (1) Install Python 3 and `pip install numpy scipy pillow matplotlib`. (2) Place the three .py files in one folder. (3) Run `python3 run_experiments.py` (about 3 minutes); all random choices use fixed seeds (text seed 7, 99 and 5; key seeds 11, 77 and 2026). (4) Run `python3 make_figs.py`. (5) Compare the printed values with the tables in this report. The same plaintexts of 100, 1,000, 10,000 and 100,000 characters (prefixes of one generated text) are used for all schemes.

Test group (decrypt(encrypt(m)) = m unless stated)	Cases	Result	Edge condition covered
Caesar, monoalphabetic, Vigenère, columnar, Hill at 100 / 1,000 / 10,000 / 100,000 characters	20	20 pass	length not a multiple of 8 or 3
Hill: PT20481, AMOX500MG, P, AB, 9	5	5 pass	digits; padding (PT20481 → PT20481XX)
Empty input; Caesar k = 0 and 26; Vigenère key "A" (identity, flagged weak)	3	3 pass	empty input, weak keys
Singular Hill key [[2,4,6],[1,3,5],[7,8,9]] rejected (gcd(det, 36) = 36)	1	1 pass	invalid key
Cleaning: "PT-20481, AMOX 500MG" → letters only	1	1 pass	punctuation, spaces, digits
Total	**30**	**30 of 30 pass**	

Expected versus actual. Hand calculation of the first Hill block of PT20481, "PT2" = (15, 19, 28): row 1 = 1·15 + 2·19 + 8·28 = 277 ≡ 25 → Z; row 2 = 5·15 + 7·19 + 9·28 = 460 ≡ 28 → 2; row 3 = 2·15 + 5·19 + 6·28 = 293 ≡ 5 → F. Program output for the whole ID: Z2F88KF9T, which begins Z2F as expected and decrypts to PT20481XX.

TASK 10 – Record and Interpret Experimental Evidence

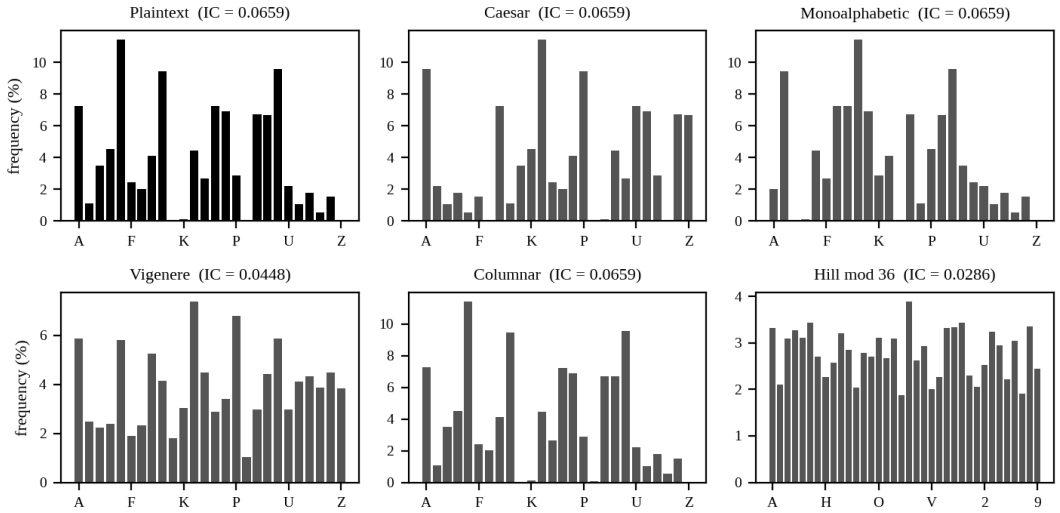


Figure 4. Letter-frequency histograms of 10,000 letters: plaintext and each ciphertext, with the index of coincidence (IC).

Scheme	IC measured	Reference value	Reading
Plaintext	0.0659	English ≈ 0.0667	Peaked histogram (E, T, A, O …).
Caesar / Monoalphabetic	0.0659	unchanged	Same peaks relabelled: frequency analysis works.
Columnar / Mono + columnar	0.0659	unchanged	Letters only moved: IC cannot see transposition.
Vigenère (m = 6)	0.0448	≈ 0.043 expected	Flatter: several alphabets mixed; IC reveals polyalphabetic cipher.
Hill mod 36	0.0286	1/36 = 0.0278	Almost flat over 36 symbols: strongest against ciphertext-only.

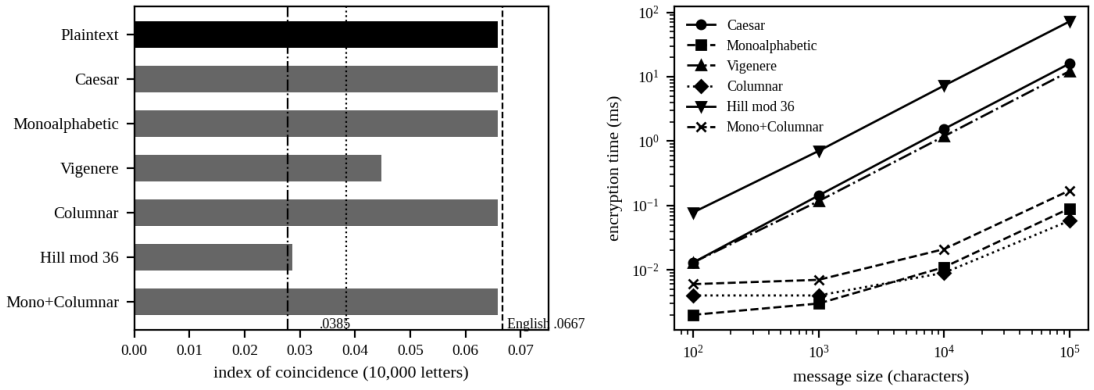


Figure 5. Index of coincidence per scheme (left) and encryption time against message size (right, log–log).

Encryption / decryption time, ms	100 chars	1,000 chars	10,000 chars	100,000 chars
Caesar	0.013 / 0.015	0.143 / 0.146	1.542 / 1.675	15.995 / 8.86
Monoalphabetic	0.002 / 0.002	0.003 / 0.003	0.011 / 0.011	0.09 / 0.09
Vigenere	0.013 / 0.013	0.117 / 0.123	1.189 / 1.27	12.171 / 13.183
Columnar	0.004 / 0.011	0.004 / 0.066	0.009 / 0.621	0.058 / 6.54
Hill mod 36	0.078 / 0.086	0.703 / 0.766	7.268 / 7.524	72.094 / 80.762
Mono+Columnar	0.006 / 0.014	0.007 / 0.075	0.021 / 0.64	0.171 / 6.69

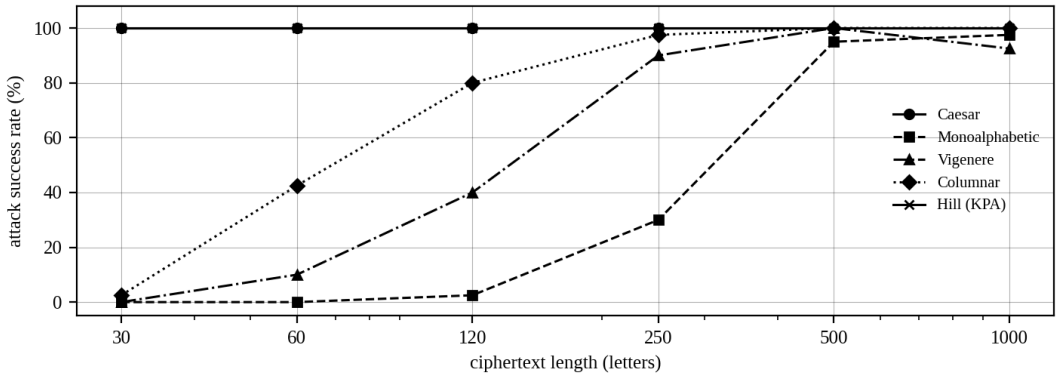


Figure 6. Attack success rate against ciphertext length (40 random trials per point).

Attack success, %	30 letters	60 letters	120 letters	250 letters	500 letters	1000 letters
Caesar (brute force)	100	100	100	100	100	100
Monoalphabetic (≥90% letters)	0	0	2.5	30	95	97.5
Vigenère (key recovered)	0	10	40	90	100	92.5
Columnar (exact text)	2.5	42.5	80	97.5	100	100
Hill (known plaintext, 3 blocks)	100	100	100	100	100	100
Mono: mean letters correct	17%	18%	45%	69%	97%	99%

Interpretation. (1) IC separates the classes exactly as the theory of Task 5 predicts: unchanged IC (0.066) for Caesar, monoalphabetic and columnar means frequency-preserving ciphers; the drop to 0.045 for Vigenère matches the mixture of 6 alphabets; the near-flat 0.029 for Hill matches 1/36. (2) The attack curves show a **length threshold**, not a key-space effect: the monoalphabetic cipher (88 bits) is no safer than Vigenère (28 bits) once about 500 letters are available, and the clinical notes of 200–10,000 characters are mostly above that threshold. (3) Hill's success rate is flat at 100% because the attack uses three blocks whatever the length; linearity, not data volume, is the weakness. (4) All times grow linearly with size and stay below 0.1 s for 100,000 characters; Hill is the slowest only because it performs nine multiplications per three characters. Performance is therefore not a constraint; key management is.

TASK 11 – Recommend and Defend the Final Design

Recommendation (Day-3 design, before the insider change). Bell-LaPadula with compartments plus break-glass and audit for access control, and a **layered product cipher**: Vigenère (12-letter key) followed by columnar transposition (8 columns), with CS1 and CS2 checksums and a message counter inside the plaintext. It was chosen because it removes both single-letter and digram structure (IC about

0.045 and digram share halved), has 71.7 key bits against 15–56 for the single ciphers, needs only a Vigenère table and a grid, and can be programmed in a few lines.

Scheme	Key bits	Ciphertext for ≥90% break	Known plaintext	Manual effort (est.)	Verdict
Caesar	4.6	under 30 letters	one letter	very low	rejected
Monoalphabetic	88.4	about 500	cribs seed the search	low	weak
Vigenère, m = 12	56.4	about 250 (m = 6)	one 12-letter crib = key	medium	weak alone
Columnar, m = 8	15.3	about 250	cribs reveal column order	medium	weak alone
Hill 3×3 mod 36	44.1	not broken ciphertext-only here	9 characters = key	high	rejected for insiders
Vigenère-12 + columnar-8	71.7	not tested end to end	broken in Task 12 (2.2 s)	high	chosen on Day 3

Residual risks, honestly stated. The design gives no integrity or authentication (checksums are forgeable, Task 7); the one shared key is a single point of failure; known headers give cribs; ciphertext can be replayed unless the counter is checked; BLP is bypassed by anyone who can decrypt. **Are classical ciphers acceptable for patient data at all? No.** Their strength is measured in minutes of computation (Tasks 6 and 12), they provide no integrity, and patient records are covered by privacy law (for example the Indian DPDP Act 2023 and HIPAA-style rules) that expect strong modern cryptography. A real deployment needs AES-256 in an authenticated mode (GCM), digital signatures (ECDSA or RSA) for every order, TLS 1.3 tunnels between clinics and hospital, a small public-key infrastructure with a certificate per clinic and per user, hardware-protected keys (smart cards), time stamps or counters against replay, and full audit logging, together with BLP plus an integrity model.

Key distribution and usability for 25 clinics. One shared key means 25 copies and total loss if any one leaks. Per-clinic keys need 25 keys at the hospital (a star network); pairwise keys between all 26 sites would need $26 \cdot 25/2 = 325$. Classical systems must distribute keys by courier or printed key cards, and every rotation repeats the effort. Non-technical staff need a one-page procedure, pre-printed key cards, a check character in every message and a 30-minute training; any software should hide the cipher completely, because manual encryption is slow and error-prone.

TASK 12 – Re-evaluate and Adapt the Design

A. Impact Analysis (2 marks)

Threat model. Before: an outsider on the public network, no key, one or a few ciphertexts. Now: an **insider** at a clinic who may hold the key or key card, has copied hundreds of ciphertexts over months, and knows the message formats. **Failed assumptions:** (1) *key secrecy at all 25 sites*: the key exists in 25 places, so the attack surface is 25 times larger; (2) *key reuse*: one key protects every message for months, so every captured message helps with all the others; (3) *unpredictability of messages*: headers "PATIENT ID", "DIAGNOSIS:" and "PRESCRIPTION:" are known plaintext at fixed positions. **Why volume of known-format messages changes the ranking:** each message is a free crib, so cryptanalysis changes from a statistical search to simple arithmetic and the likelihood of an insider breaking the cipher rises from 2 to 5. Row 7 of the Task 1 table therefore moves to the top (25), and rows 1–3 rise because everything an insider can decrypt he can also forge.

B. Evaluation of the Original Design

Test of the recommended scheme. I generated 200 messages of 240 letters, each beginning with PATIENTID, DIAGNOSIS or PRESCRIPTION, and encrypted them under one Vigenère key (12 letters) and one columnar order (8 columns). The attacker knows only the three headers. For each of the 8! = 40,320 possible column orders he computes the Vigenère key letters implied by the nine known header letters and checks whether all messages agree; the true order is the only one where every message gives the same key letters.

Measure	Result
Consistency score of the correct column order (maximum 9.00)	9.00 of 9.00; recovered in 2 s
Key recovered (header PRESCRIPTION gives all 12 letters)	ZIKGHGDJPRTH = true key ZIKGHGDJPRTH
Messages decrypted with the recovered material	200 of 200
Smallest set tested (3, 6, 12, 30 captured messages)	column order recovered in every case
Attack effort	$\approx 2^{18.5}$ operations, against $2^{71.7}$ for brute force of the 71.7-bit key
Same attack when every message uses its own key and order (redesign)	0.68 of 9.00 = chance level: no key, no order

Where BLP still holds and where it is irrelevant. BLP still holds for every request that passes through the hospital's reference monitor: it stops a nurse from reading a Secret record via the application and a clerk from reading lab results (Task 3). It is **irrelevant against an insider who holds the key**, because decryption happens outside the monitor: with one key for all levels, a Confidential-level key opens Secret diagnoses, the star property is never consulted, and the insider can also forge valid ciphertext. Cryptographic keys must therefore follow the BLP labels (a different key per level and compartment), otherwise the access policy and the encryption policy disagree.

C. Adaptation or Redesign

Decision: REDESIGN. Retaining the design is impossible because it is broken in seconds. Modifying it (longer key, more rotation) does not help while cribs and a shared key remain. The redesigned flow is shown in Fig. 7.

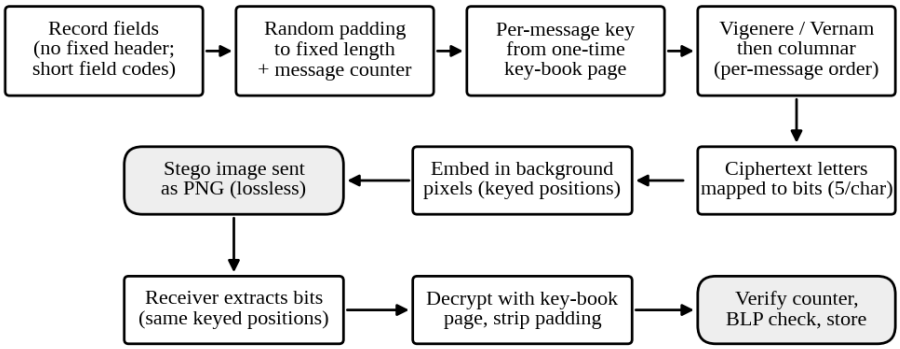


Figure 7. Redesigned flow: no fixed header, random padding, per-message key from a one-time key book, per-message column order, then encrypt-then-embed in a lossless PNG scan.

What changes and its effect on key space and attack effort. (1) *Per-message key from a one-time key book* (a Vernam-style pad over mod 26, one page per message, destroyed after use, one book per clinic, monthly rotation): the key length equals the message length, so the key space becomes 26^L (for 240 letters about 2^{1128}) instead of 2^{56}; a crib reveals only that message's key stream and nothing about any other (measured 0.68 of 9.00). (2) *No fixed headers*: short field codes inside the encrypted body, random-length padding and a random prefix remove the fixed crib position (an attacker must try each offset, a modest ×12 factor, but with per-message keys it gains nothing). (3) *Per-message column order* taken from the pad page adds 8! orders per message instead of one global order. (4) *Per-clinic books* mean one insider exposes only his clinic's pages. (5) *Integrity* cannot be added classically, so a message counter, callback confirmation of every dose change by a second person, and the checksum (as a detector of accidents only) are used procedurally. (6) *Scan images*: encrypt first, then embed in background pixels of a PNG (Task 8). Cost: a clinic that sends 200 messages of 240 letters a month needs about 48,000 pad letters, roughly 24 printed pages (my assumption of 2,000 letters per page), delivered by courier.

D. Evidence-Based Defence

Criterion	Original (shared Vigenère + columnar)	Redesigned (per-message pad, padding, encrypt-then-embed)	Modern production design
Key space per message	$2^{71.7}$, one key for all	26^L per message (240 letters ≈ 2^{1128}) × 8!	2^{256} (AES-256)
Crib / known-plaintext attack	$2^{18.5}$ work; all 200 messages opened	per message only; 0.68 of 9.00	no practical attack
Insider with one clinic key	reads and forges for all 25 clinics	reads only that clinic's pad pages	only that session or user
Integrity and authentication	none (checksum forgeable)	procedural: counter, callback	AEAD tag plus digital signature
Usability and cost	cheap, simple, manual	monthly courier of pad books; one-day training	automatic software; smart cards
Verdict	broken	acceptable only for training or low-risk legacy use	required for patient data

Final defence. My mathematics shows that key size does not equal strength (88 bits for the monoalphabetic cipher falling to 500 letters); my experiments show the same effect for reuse (the shared 12-letter key opened all 200 messages in 2.2 s); the insider threat shows that the secrecy of one key at 25 sites cannot be assumed; and the usability analysis shows that the only classical design that resists the insider (a one-time pad) costs a courier and a printed book per clinic every month. **What classical techniques can never guarantee for patient data:** integrity of a dosage, authentication of the doctor, non-repudiation of an order, protection against a reused or leaked key, forward secrecy, secure key distribution without a courier, and availability. They are valuable teaching tools, which is how this assignment has used them, but a real hospital network must use authenticated modern encryption with digital signatures, with Bell-LaPadula (plus an integrity model) as the access-control layer on top.

SIMATS ENGINEERING

CSA5116 – Cryptography and Network Security

ASSIGNMENT 1 – CO1

Design, Analyse and Validate Classical Encryption Techniques for Securing Sensitive Information

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.

Student Name	Register Number	Department / Section	Academic Year / Semester	Faculty Name	Nature of Assignment	Duration	Total Marks	Date of Submission	Date of Evaluation
		CSE	2026–2027	Dr. D. Sudhagar	Individual	1 Week	50		

MARK SHEET (Rubric-based Evaluation)

Task	Task Title	Max	Marks Awarded	Remarks / Evidence Expected
1	Cipher Design and Selection	5		Select and describe three suitable classical ciphers (Caesar, monoalphabetic substitution, and Vigenère or Hill); justify their use and limitations for the given sensitive message.
2	Encryption and Decryption	5		Show correct key selection, encryption steps, ciphertext, decryption steps and recovery of the original plaintext.
3	Key Analysis and Mathematical Basis	5		Explain key space, modular arithmetic or alphabet mapping used by each selected cipher; calculations must be traceable.
4	Security and Cryptanalysis	5		Analyse vulnerabilities such as brute-force search, frequency analysis, repeated-key patterns and known-plaintext attacks where relevant.
5	Comparison of Cipher Strength	4		Compare the selected methods using key space, implementation effort, likely attacks and suitability for sensitive information.
6	Algorithm / Pseudocode Formulation	4		Provide clear encryption and decryption pseudocode or flowcharts, including input validation and treatment of spaces/punctuation.
7	Worked Example and Verification	4		Use a sample message; verify that decrypt(encrypt(plaintext)) reproduces the original message under the stated rules.
8	Implementation and Testing	4		Implement at least two ciphers in a programming language; include source code, test cases, expected output and actual output.
9	Experimental Evaluation	3		Test multiple messages and key choices; present organised tables of ciphertext, recovered plaintext, execution time or operation counts.
10	Theory-to-Result Discussion	3		Explain how observed results match the cipher design and why classical ciphers are not suitable for modern high-security use.
11	Recommend and Defend a Design	4		Recommend a method for the assignment scenario and defend the decision using correctness, usability and security limitations.
12A	Re-evaluate: Impact Analysis	2		Discuss the impact of message length, key choice and repeated patterns on confidentiality.
12B	Re-evaluate: Attack Evaluation	2		Evaluate likely attacks against the selected ciphers and identify information an attacker could exploit.
12C	Re-evaluate: Adaptation or Redesign	2		Propose a safer modern replacement or layered approach, explaining why classical ciphers are only educational examples.
12D	Re-evaluate: Evidence-Based Defence	2		Defend conclusions using worked examples, test results, key-space reasoning and a comparison table.
TOTAL		50		Marks to be awarded by the faculty.

OVERALL REMARKS

Faculty feedback: _____

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Student Signature	Faculty Signature	Program Director
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MANDATORY SUBMISSION EVIDENCE CHECK

✓ = present P = partly present X = missing. Mark the status after checking the student's submitted report and supporting evidence.

Task	Evidence Expected	Status	Note
1	Cipher selection, design rationale and assumptions	☐	Three ciphers described and justified
2	Encryption and decryption working shown	☐	Steps, key and plaintext/ciphertext included
3	Mathematical basis and key-space explanation	☐	Calculations are traceable
4	Security analysis and likely attacks	☐	Relevant weaknesses identified
5	Cipher comparison table	☐	Criteria applied consistently
6	Pseudocode or flowchart	☐	Encryption and decryption steps included
7	Worked example and correctness verification	☐	Recovered plaintext matches original
8	Source code and reproducible tests	☐	Program, test inputs and outputs included
9	Experimental results in tables	☐	Multiple inputs or keys evaluated
10	Theory and experimental results compared	☐	Observations interpreted
11	Recommendation supported by analysis	☐	Trade-offs and limitations stated
12A–D	Impact, attack evaluation, redesign and evidence-based defence	☐	All four parts addressed

IMPLEMENTATION EVIDENCE CHAIN (Tasks 8–9)

Evidence Item	Status	Observation
Source code	☐	Readable source code for each implemented cipher; language and execution instructions stated
Test inputs	☐	Plaintexts, keys and edge cases recorded
Expected behaviour	☐	Expected ciphertext and recovered plaintext documented
Actual output	☐	Program output or screenshots included
Intermediate evidence	☐	Key mapping, modular steps or intermediate transformations shown
Boundary / failure tests	☐	Empty input, spaces, punctuation, invalid keys and repeated characters considered
Interpretation	☐	Results connected to cipher properties and security limitations
Code, data and report consistent	☐	Tables and claims match the submitted program output

SUBMISSION CHECKLIST (not separately marked)

Requirement	Status
All assignment tasks including 12A–12D answered	☐ Yes ☐ Partly ☐ No
Encryption and decryption shown for each selected cipher	☐ Yes ☐ Partly ☐ No
Keys and assumptions clearly stated	☐ Yes ☐ Partly ☐ No
At least two ciphers implemented and tested	☐ Yes ☐ Partly ☐ No
Same plaintext conventions used consistently	☐ Yes ☐ Partly ☐ No
Source code and reproducible test procedure provided	☐ Yes ☐ Partly ☐ No
Tables / figures / program outputs included	☐ Yes ☐ Partly ☐ No
Cross-references and numbering correct	☐ Yes ☐ Partly ☐ No
No unfilled template placeholders in the student report	☐ Yes ☐ Partly ☐ No
Viva recommended where implementation or calculations need clarification	☐ Yes ☐ No